Student Grade Management System

Data Dictionary

CiT Backend Engineering Track — Week 6: Databases & SQL Foundations

1. Purpose

This document describes the structure of the relational database supporting the Student Grade Management System. It records every table, column, data type, constraint, and relationship used to persist student records and their subject scores in PostgreSQL.

2. Database Overview

Database name: `cit_school`

Engine: PostgreSQL 18

Tables: students, scores

3. Table: students

Stores one record per student, including their name and unique registration number.

Table Structure

Column	Data Type	Nullable	Key	Description
`id`	`SERIAL`	No	Primary Key	Auto-incrementing unique identifier for each student record.
`name`	`VARCHAR(100)`	No	—	Full name of the student.
`registration_number`	`VARCHAR(20)`	No	Unique	Unique registration number assigned to the student. Used as the lookup key for the system's Find and Add Score operations.

4. Table: scores

Stores one record per subject score. Each row belongs to exactly one student, allowing a student to hold any number of scores without repeating their personal details.

Table Structure

Column	Data Type	Nullable	Key	Description
`id`	`SERIAL`	No	Primary Key	Auto-incrementing unique identifier for each score record.
`student_id`	`INT`	No	Foreign Key	References students(id). Identifies which student this score belongs to.
`score`	`INT`	No	Check (0–100)	Numeric score for one subject. Constrained to the range 0–100 at the database level.

5. Relationships

- **students (1) → (many) scores:** a single student can have many score entries, but each score entry belongs to exactly one student.
- **Enforced by:** `scores.student_id REFERENCES students(id)`

6. Constraints Summary

Constraint	Purpose
PRIMARY KEY	Uniquely identifies each row in students and scores.
NOT NULL	Ensures name, registration_number, student_id, and score can never be left empty.
UNIQUE	Prevents two students from sharing the same registration_number.
FOREIGN KEY	Ensures every score row points to a student that actually exists.
CHECK	Restricts score values to the valid range of 0 to 100.

7. Source Definition (DDL)

```
CREATE TABLE students (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    registration_number VARCHAR(20) UNIQUE NOT NULL
);

CREATE TABLE scores (
    id SERIAL PRIMARY KEY,
    student_id INT NOT NULL REFERENCES students(id),
    score INT NOT NULL CHECK (score >= 0 AND score <= 100)
);
```